

Troubleshooting Guide

Problem 1 — VLAN Not Working

Symptoms

A PC cannot communicate with other devices in its VLAN.

Checks

show vlan brief

Verify that the switch port belongs to the correct VLAN.

Solution

```
interface fastethernet 0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

Problem 2 — Inter-VLAN Routing Not Working

Symptoms

Devices can communicate within their VLAN but cannot communicate with another VLAN.

Checks

show ip interface brief

Verify router subinterfaces.

Example:

```
interface gigabitethernet 0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

Problem 3 — Trunk Failure

Symptoms

VLAN traffic does not reach the router.

Check

show interfaces trunk

Solution

interface gigabitethernet 0/1

switchport mode trunk

Problem 4 — Incorrect Gateway

Symptoms

The PC cannot communicate outside its local subnet.

Check

Verify the PC's default gateway.

Example:

Admin PC:

192.168.10.10

Gateway:

192.168.10.1

Problem 5 — ACL Blocking Unexpected Traffic

Check

show access-lists

Review the ACL rules and packet counters.

Ensure the ACL is applied to the correct interface and direction.

Troubleshooting Methodology

The following order can be used:

Check physical connections



Check IP addresses



Check VLAN assignment



Check trunk configuration



Check router subinterfaces



Check routing



Check ACL configuration



Perform connectivity tests

This structured process helps isolate networking problems systematically.